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Symptom-based depression subtypes: brain dynamic specificity and its association with gene expression profiles

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At least 227 combinations of symptoms meet the criteria for Major Depressive Disorder (MDD). However, in clinical practice, patients consistently present symptoms in a regular rather than random manner, and the neural basis underlying the MDD subtypes remains unclear. To help clarify the neural basis, patients with MDD were clustered by symptom combinations to investigate the neural underpinning of each subtype using functional resonance imaging (fMRI). Four symptom-based subtypes of MDD were identified using latent profile analysis according to the clinical scales. Subsequently, brain dynamics were evaluated using fMRI, and the dysregulations in attention and limbic network were observed among the subtypes. Correlation between brain dynamics and symptom combinations was then assessed via canonical correlation analysis (CCA). The brain-symptom correlation was higher when evaluated in subtypes ($r = 0.77$ to 0.92) compared to the entire group ($r = 0.5$). The loading weight in CCA showed that dynamics in transmodal networks contributed the most to the correlation in the subtypes characterized by typical depression symptoms, whereas unimodal networks contributed the most to subtypes characterized by anxiety and insomnia. Finally, gene expression underlying the CCA model, along with its biological encoding process, performed using a postmortem gene expression atlas revealed distinct gene enrichments for different subtypes. These findings highlight that distinct symptom clusters in MDD have specific neural correlates, providing insights into depression's heterogeneous diagnosis and precision medicine opportunities.

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INTRODUCTION

Major depressive disorder (MDD) is a prominent public health concern [1] with a substantial financial burden [2] that is compounded by the escalating risk of suicide among young patients with depression [3]. Clinically, MDD is a heterogeneous disorder, a challenge that may impede a deeper understanding of its neural underpinnings [4] and hinder the attainment of consensus, even in meta-analyses [5].

Clarifying the phenotypes of MDD represents a promising approach to addressing its heterogeneity [6]. Clinically relevant subtypes in MDD were recognized decades ago [7]. For instance, anxious depression was identified as a subtype characterized by severe anxiety among patients in the latter half of the twentieth century [8, 9]. More recently, Ulbricht et al. [10] categorized male patients into four subtypes based on the patterns of symptoms regarding depression severity and psychomotor disturbance. Nonetheless, the insufficient focus on the biological underpinnings of these symptom-based subtypes hinders their widespread application in clinical practice.

Study of brain dynamics, which refer to the temporal patterns of brain activity, is crucial for elucidating the pathophysiology of

MDD. Using functional magnetic resonance imaging (fMRI), studies have demonstrated alterations in dynamic functional connectivity in MDD [11–14] and have associated these changes with symptom severity [15, 16]. However, the results of a meta-analysis have indicated that the heterogeneity of MDD impedes the drawing of consistent conclusions from dynamic fMRI studies [17]. Nevertheless, there is evidence that dynamic features can differentiate psychiatric conditions [18] and enhance the performance of machine learning models in classifying biotypes of attention deficit hyperactivity disorder [19]. Such evidence suggests that dynamic features are sensitive in identifying distinct neural patterns among subgroups of mental illnesses. Collectively, these facts highlight the importance of considering brain dynamics in the subtyping of MDD, thereby deepening our comprehension of the neural underpinnings of this disorder.

In the current study, our objective was to reveal the neural dynamic signatures associated with symptom-based subgroups of depression. We performed a four-step process. First, we employed latent profile analysis to categorize patients into subtypes based on their symptomatic patterns. Second, we compared whole-brain dynamic patterns among these subtypes and contrasted them

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with healthy controls (HCs). We assessed the whole-brain dynamic patterns using time delay (TD) estimation, a method of identifying the stereotyped propagation patterns throughout the brain [20, 21]. Third, we obtained correlation pattern maps between symptoms and brain dynamics using canonical correlation analysis (CCA), illustrating the specific brain-symptom association for each subtype. Our final step was based on previous studies revealing that the specific brain dynamic pattern was driven by a unique gene profile [22, 23]. Based on this discovery, we investigated the underlying biological processes of gene expression linked to the subtype-specific brain-symptom correlation maps by leveraging gene expression patterns from the Allen Human Brain Atlas [24].

MATERIALS AND METHODS

Participants

A total of 184 patients were recruited from the Department of Depression in Shenzhen Kangning Hospital. The inclusion criteria were (1) age 18 years or older; (2) diagnosis of MDD according to the Diagnostic and Statistical Manual of Mental Illnesses Fifth Edition (DSM-5) criteria; (3) absence of comorbid mental disorders; (4) a Hamilton Depression Rating Scale (HAM-D) total score of at least 17; and (5) no contraindications to MRI scanning. Clinical scales, including the HAMD and the Hamilton Anxiety Rating Scale (HAMA), were evaluated during the first week of the patient's hospitalization. One hundred adult HCs were recruited from the local community. Data from 49 patients and 9 HCs were excluded due to excessive head motion during fMRI scanning. Head motion was assessed using frame-wise displacement (FD), with mean FD values exceeding 0.25 or the occurrence of spike volumes exceeding 25% of the time series indicating excessive movement. Consequently, data from 135 patients and 91 HCs were included in the formal analysis.

Determining subtypes by latent profile analysis

Latent profile analysis (LPA) was applied to define the patient subtypes based on the item-level scores of the HAMD. Given that HAMD corresponds to a specific clinical symptom, clustering based on HAMD items enables the identification of symptom-specific subtypes among patients. LPA uses a set of variables to discern latent subpopulations within a larger population and assumes that people can be categorized to K latent profiles according to different configurable profiles of their personal characteristics [25] based on the following formula:

$$\delta_i^2 = \sum_{k=1}^K \pi_k (\mu_{ik} - \mu_i)^2 + \sum_{k=1}^K \pi_k \delta_{ik}^2$$

where μ_{ik} and δ_{ik} are profile-specific (k) means and variances for HAMD item i . π_k represents the proportion of participants that belong to profile k .

Next, an exploration to determine the optimal number of categories, ranging from 2 to 6, was conducted. The determination of the optimal category model was based on application of the analytic hierarchy process (AHP) with five model fitness, including Akaike's information criterion (AIC), Bayesian information criterion (BIC), approximate weight of evidence (AWE), classification likelihood criterion (CLC), and kullback information criterion (KIC) [26]. The model selected as optimal by the AHP was subsequently reviewed by our clinical team to confirm its clinical applicability. LPA was using the R package tidyLPA [27].

MRI data acquisition

MRI data were obtained using a 3.0 Tesla scanner (Discovery MR750 System; General Electric, Milwaukee, WI, USA) with an eight-channel head coil at Shenzhen Kangning Hospital. All participants were instructed to remain still and awake with their eyes closed and not to think about anything. The fMRI data were obtained using an echo-planar imaging (EPI) sequence with the following parameters: repetition time (TR) = 2000 ms, echo time (TE) = 30 ms, flip angle = 90°, thickness/gap = 3.5/0.7 mm, acquisition matrix = 64 × 64, field of view (FOV) = 224 mm², 33 axial slices, and 240 time points (8 min). In addition, high-resolution structural MRI scans were obtained using a fast-field echo (FFE) three-dimensional T1-weighted (3D-T1WI) sequence. The detailed parameters were as follows: TR/TE = 6.65/2.93 ms, flip angle = 12°, acquisition matrix = 256 × 256, FOV = 256 × 256 mm², and 192 sagittal slices with no inter-slice gap.

Functional data preprocessing

We performed quality control (QC) on the images using MRIQC ver. 22.0.6 before image preprocessing. Subsequently, the functional images were preprocessed via fMRIPrep ver. 22.1.1 [28]. Briefly, the steps included (1) removing the first 10 volumes; (2) performing slice-timing correction; (3) performing head-motion correction with six parameters; (4) performing alignment between functional images to T1w images; and (5) performing spatial normalization to the Montreal Neurological Institute (MNI) standard space. After the images were preprocessed, time series of cortical signals were extracted over a Schaefer-400 atlas [29] and the subcortical signals over the Harvard-Oxford subcortical structural atlas (14 regions), regressing out head-motion parameters, white matter, and cerebrospinal fluid signal. Notably, for the head-motion parameters, we employed the 24-parameter model [30], which includes rigid body parameters, the derivatives, quadratic terms and power2d derivatives. After that, a linear detrending, band-pass for 0.01 to 0.1 Hz and smoothing for 6 mm full-width half-maximum kernel were performed. The regression for the global signal was not performed to ensure the precision of calculating the dynamic metric [31]. Nuisance regression and blood-oxygen level dependent (BOLD) signal extraction was performed using the Python package Nilearn (<https://nilearn.github.io/dev/index.html>).

Time delay estimation

The propagation of the BOLD signal throughout the brain follows a temporally regular sequence rather than occurring randomly [31]. Time delay (TD) is proposed to measure the relative lag structure between brain areas in this propagation sequence [20]. Previous studies have shown that the shifts in TD are associated with the progression of MDD [32] and neurological diseases [33–35], suggesting that TD can reveal aberrant dynamic structures in psychiatric conditions. Therefore, TD was estimated to assess the dynamics of brain activity.

The computation of TD is based on the cross-covariance function (CCF) between two time series $x_1(t)$ and $x_2(t)$. In practice, CCF was applied to the shifted time series as

$$c_{x_1 x_2}(\Delta) = x_1(t+\Delta) \cdot x_2(t)$$

where Δ is the shifted interval corresponding to the TR units such as [TR-2, TR-1, TR, TR+1, TR+2]. Time delay ($\hat{\tau}_{1,2}$) is then estimated by a three-point parabolic interpolation on the peak of $c_{x_1 x_2}(\Delta)$ as

$$\hat{\tau}_{1,2} = TR \frac{c_{peak-1} - c_{peak+1}}{2(c_{peak-1} - 2c_{peak} + c_{peak+1})}$$

where c_{peak} is the peak among $c_{x_1 x_2}(\Delta)$ and the values immediately preceding (c_{peak-1}) and succeeding (c_{peak+1}). The above approach was applied to each pair of the parcel, resulting in a 414 × 414 TD matrix. Then, a time delay projection map was generated by column-wise averaging of the TD matrix, resulting in a vector with 414 elements. Finally, the TD of somatomotor network (SMN) was extracted by averaging the SMN-related elements in the time delay projection map. TD estimation was performed with inhouse scripts, with adjustment to the script provided by Raut, Mitra, Snyder, and Raichle [36].

Brain-symptom canonical correlation analysis

After the subtypes were determined, canonical correlation analysis (CCA) was conducted to obtain the symptom-brain correlation maps ($N_{\text{subject}} \times 17$) for each cluster of patients using the item-level HAMD scores and cerebral cortex time delayed projection map ($N_{\text{subject}} \times 400$). The subcortical regions were not included in the CCA and the following gene-wise analysis given the large difference of transcription between cortical and subcortical regions [37]. CCA finds the linear combinations of the input matrices X and Y , as

$$U^1 = Xa^1 = a_1^1 X^1 + a_2^1 X^2 + \dots + a_p^1 X^p$$

and

$$V^1 = Yb^1 = b_1^1 Y^1 + b_2^1 Y^2 + \dots + b_q^1 Y^q$$

where p and q are the dimensions of X and Y . Therefore, the correlation ecoefficiency ρ_1 was obtained by

$$\rho_1 = \text{cor}(U^1, V^1)$$

Table 1. Demographic data for participants.

	HC (N = 94)	saMDD (N = 28)	sdMDD (N = 48)	miMDD (N = 27)	mdMDD (N = 32)
Gender					
female	33 (35.1%)	17 (60.7%)	40 (83.3%)	26 (81.3%)	17 (63%)
male	61 (64.9%)	11 (39.3%)	8 (16.7%)	6 (18.8%)	10 (37%)
Age					
Mean (SD)	34.4 (11.3)	37.8 (10.8)	31.9 (13.0)	31.9 (14.5)	33.7 (12.9)
HAMD					
Mean (SD)	\	32.2 (4.34)	30.4 (3.91)	22.4 (3.89)	20.7 (2.84)
Median [Min, Max]	\	32 [25, 41]	30 [24, 41]	21 [17, 32]	21 [17, 27]

In the parenthesis, the percentage represents the proportion out of the total for a specific column.

HAMD hamilton depression rating scale, SD standard deviation, saMDD severe anxious MDD, sdMDD severe depressive MDD, miMDD moderated insomnia MDD, mdMDD moderated depressive MDD.

Given that the number of patients in each subtype was far less than the dimensions of time-delayed estimation, regularized CCA (rCCA) was used instead of traditional CCA to prevent model overfitting [38]. rCCA is widely applied in neuroimaging and gene studies [39, 40] to solve the dimension-subject conflicts in these fields. rCCA introduces a regularization step in CCA by adding ridge penalties (λ_1, λ_2) to the diagonal of X and Y . The optimal combination of (λ_1, λ_2) that maximizes the CCA ecoefficiency was determined through leave-one-out cross-validation among a predefined searching grid (see González et al. [41] for more details). In the present study, the best fit (λ_1, λ_2) was searched for within a 51×51 grid.

rCCA was conducted for every LPA cluster separately, and the rCCA ecoefficiency was obtained to evaluate the subtype-specific strength of the correlation between TD estimation and symptoms according to HAMD. Sequentially, 5000-time permutation tests were performed to examine whether the rCCA coefficient obtained for each MDD subtype was greater than that obtained by chance. As a control, rCCA was also conducted on the entire cohort of the patients to investigate whether patient heterogeneity attenuates the brain-symptom correlation. Additionally, the first component of CCA coefficient for each brain parcel was extracted to create an rCCA loading map, which illustrates the parcel-specific contributions to the brain-symptom relationship. rCCA was conducted in R using the CCA package [41].

Association between gene expression and symptom-brain correlation map

To bridge the macro- and microscopic phenotypes of MDD subtypes, the association between gene expression and the CCA maps was investigated. For this aim, regional microarray expression data of the left and right hemispheres were obtained from the Allen Human Brain Atlas (AHBA, <https://human.brain-map.org>; Hawrylycz, et al. [24]), which were obtained from six post-mortem brains (One female, ages 42.50 $\pm$ 13.38). The gene profile was processed using the abagen toolbox (<https://github.com/rmarkello/abagen>), aligning it with the Schaefer-400 atlas in MNI space (see the supplementary materials for further details). Due to the sampling bias of the right hemisphere in the AHBA, the robustness of the results was assessed by using gene expression data obtained through mirroring the left hemisphere onto the right side, a method proposed to mitigate lateralization bias [42].

Four partial least squares (PLS) regressions were conducted to explore the association between the transcriptional profile and the rCCA map for each subtype. In this analysis, the gene expression data and the CCA map were set as the predictor and response variables, respectively. Genes were then ranked based on their contribution to the component of the PLS model that explained the greatest variance in gene expression. A spatial autocorrelation corrected permutation test [43] was performed to examine whether the variance explanation (R^2) of the PLS component was greater than that expected by chance. Finally, the gene sequences were then subjected the gene enrichment analysis to identify enriched Gene Ontology terms of the biological process using Gorilla (<http://cbl-gorilla.cs.technion.ac.il/>) [44]. Significant enrichment was defined as $P < 1 \times 10^{-5}$.

Statistical analysis

Demographics and clinical scales among subtypes. Differences in age and gender among the MDD subtypes were examined using one-way analysis of variance (ANOVA) and the chi-square test. To assess the severity of depression and anxiety among different subtypes of patients, one-way ANOVA was performed to examine between-group differences in total scores HAMD and HAMA scores. Subsequently, pairwise comparisons were conducted to determine the effects between each pair of subtypes. Statistical significance was defined as $P < 0.05$ after applying false discovery rate (FDR) correction.

Temporal shifts among subtypes. To illustrate changes in the brain dynamic for MDD subtypes, the network-specific differences in TD estimation was compared across all groups. The participants' TD projection map was averaged into eight networks according to Yeo's seven-network atlas [45] and a subcortical network. Two-way ANOVA was performed to identify TD differences among the five groups within these eight networks, with age and gender as covariates. Post-hoc analyses were conducted to identify pairwise group differences within each network, with statistical significance set as $P < 0.05$ after applying FDR correction.

Coherence between gene correlation and dynamic aberrance. To further evaluate the plausibility of the PLS model, the consistency of the gene correlation pattern with the brain aberrant pattern was tested. Specifically, Pearson's correlation between the PLS score of the brain regions and the HC-contrast TD shifts was determined to test the consistency, with the statistical threshold of the Pearson's correlation defined as $P < 0.05$. All statistical analyses were performed using R base and the bruceR [46] packages.

RESULTS

Clinical profile of MDD subtypes

Table 1 and Fig. 1 illustrate the clinical profile of the four MDD subtypes identified by LPA (the results of the modal comparisons are shown in Table S1). No significant difference in age and gender was found among the subtypes (age: $F(3,131) = 1.512$, $P = 0.214$; sex: $\chi^2 = 7.339$, degrees of freedom [df] = 3, $P = 0.061$). Among these, two clusters emerged in the severe MDD group based on their mean HAM-D score, whereas the other two clusters were classified as moderate MDD (Fig. 1B). Within the severe MDD clusters, we distinguished severe anxious MDD (saMDD) and severe depression MDD (sdMDD) groups. The saMDD group exhibited the highest scores for agitation (item 9) and anxiety-related symptoms (items 10 and 11), whereas the sdMDD demonstrated the highest levels of depressed mood (item 1), feelings of guilt (item 2), suicidality (item 3), and loss of interest (item 7). Among the moderate MDD clusters, we identified moderate depression MDD (mdMDD) and moderate insomnia MDD (miMDD). Notably, the mdMDD group

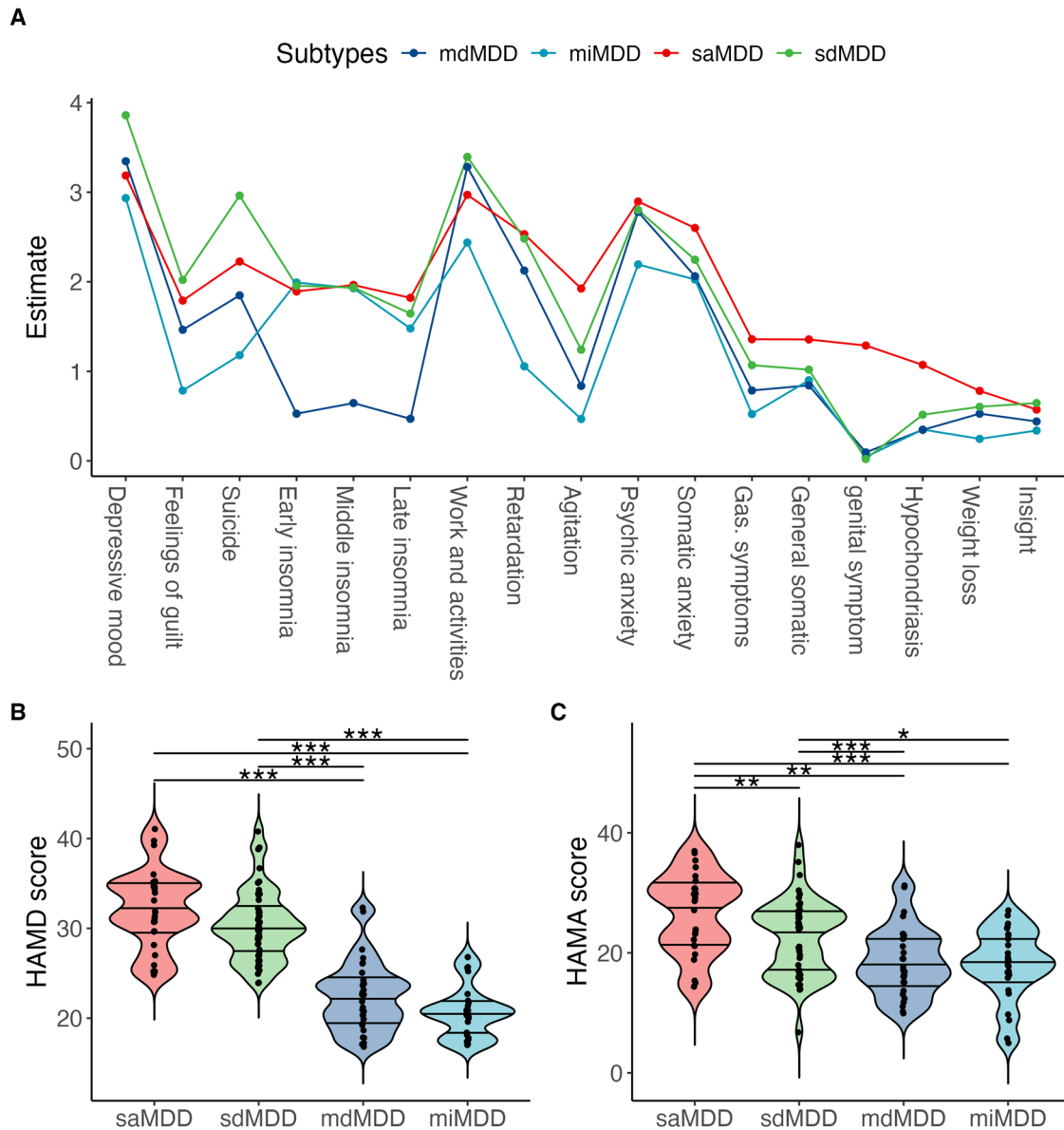


Fig. 1 Subtype identification and clinical profile. **A** Score on HAMD-17 items for each subtype. **B** HAMD total score comparisons across subtypes. **C** HAMA total score comparisons across subtypes. Abbreviations: aMDD severe anxious MDD, sdMDD severe depressive MDD, miMDD moderated insomnia MDD, mdMDD moderated depressive MDD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

showed a high level of classical depressive symptoms but a lower level of insomnia compared with the miMDD group (Fig. 1A). Additionally, the HAM-A scores indicated the highest anxiety levels in the saMDD group compared with the other three subtypes (Fig. 1C).

Lag structure difference across groups

Figure 2A shows the whole-brain map of the HC-contrast effect size for each subtype. The results of two-way ANOVA revealed a significant interaction effect of group $\times$ network ($F(28,1554) = 1.579$, $P = 0.028$). Simple effect analysis identified significant group differences in the dorsal attention network (DAN) ($F(4,222) = 2.783$, $P = 0.028$) and limbic network ($F(4,222) = 2.906$, $P = 0.023$). Pair-wise comparison showed that TD of DAN was late in the miMDD group compared with the HCs ($t = 3.286$, $P = 0.012$, Fig. 2B) and the TD of the limbic network was late in the sdMDD group compared with than two mild symptom subgroups (sdMDD vs. miMDD: $t = 2.769$, $pFDR = 0.042$; sdMDD vs. mdMDD: $t = 2.661$, $P = 0.042$, Fig. 2C).

Correlation between brain dynamics and symptoms

We found that the rCCA coefficient for all the MDD patients was relatively low ($\beta = 0.5$). However, upon stratifying patients into severe depression (sdMDD and saMDD) and moderated depression (mdMDD and miMDD) groups, the rCCA coefficient increased to 0.94 for patients with severe depression and 0.68 for patients with moderated depression. Similarly, the rCCA coefficient increased from 0.77 to 0.92 among the four subtypes (Fig. 3A), indicating the variance explanation from 59.29 to 82.81% of the rCCA models. The permutation test results illustrated in Fig. S2 indicate that the rCCA coefficients for the severe subtypes were significantly greater than they would be by chance (saMDD: $P_{\text{permute}} = 0.019$; sdMDD: $P_{\text{permute}} = 0.003$). In contrast, the rCCA coefficients for the moderate subtypes did not differ significantly from chance (mdMDD: $P_{\text{permute}} = 0.206$; miMDD: $P_{\text{permute}} = 0.409$). Nevertheless, the substantial magnitude of the rCCA coefficients for the moderate subtypes (mdMDD and miMDD) indicates that the brain-symptom correlations in these subtypes retain clinical relevance.

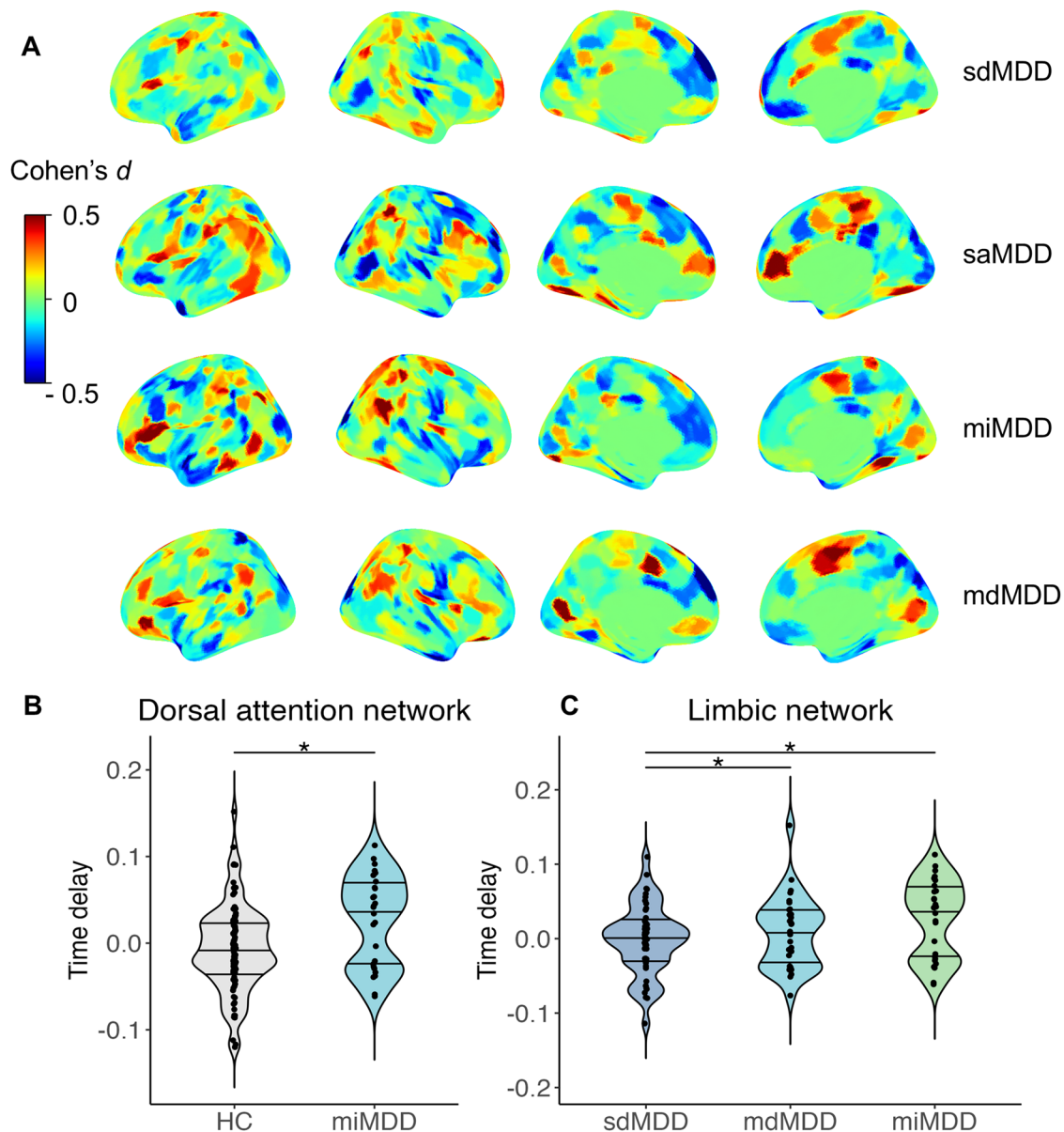


Fig. 2 Temporal shifts of brain activity in MDD subtypes. **A** Group difference of time delay projection map for HC contrast in each subtype. The effect size (Cohen's d) of each parcel is illustrated on the brain surface. **B**, **C** Violin plots show the differences in time delay estimation in dorsal attention network between miMDD and HCs and in limbic network among MDD subtypes. HC healthy control, miMDD moderated insomnia major depressive depression, MDD major depressive disorder. $*P < 0.05$.

To compare the CCA loading pattern across subtypes, we extracted and normalized the loading weights of the CCA model for the symptoms and brain areas, respectively. From the symptom loading weight results illustrated in Table S2 and Fig. S3, we observed a predominant contribution of depression-related symptoms on the CCA model in the sdMDD and mdMDD groups and anxiety-related symptoms in the saMDD group. The somatic-related symptoms exhibited the highest contribution to the miMDD group. These results suggest a convergence between the CCA pattern and the clinical profile of the subtypes.

The rCCA loading weight of the brain areas showed distinguish patterns among the MDD subtypes (Figs. 3B and S1 for the search grid result). Specifically, the orbitofrontal cortex significantly contributed to rCCA loading in the sdMDD subtype, whereas the right postcentral cortex played a crucial role in the saMDD subtype. After averaging the rCCA loadings into Yeo's seven functional networks, we found that the network-level loading

pattern differed among the MDD subtypes, as illustrated in Fig. 3C. Specifically, the limbic network had the highest rCCA loading for two severe subtypes whereas the visual network had the highest rCCA loading in two moderated subtypes. The rCCA loading in transmodal networks, including the frontoparietal network and the default mode network, was higher in the sdMDD and mdMDD subtypes than in the unimodal networks, including in the attention and primary sensory networks, whereas the opposite was observed in the saMDD and miMDD subtypes.

Gene expression profile related to brain-symptom correlation

The permutation tests revealed a higher variance explanation of the PLS model for saMDD ($R^2 > 100\%$ permuted values) and mdMDD ($R^2 > 96.7\%$ permuted values), but a lower R^2 for sdMDD ($R^2 > 62\%$ permuted values) and miMDD ($R^2 > 9.8\%$ permuted values). The PLS model revealed that the component explains 10.4% of gene expression variance in the saMDD, 11.5% in sdMDD,

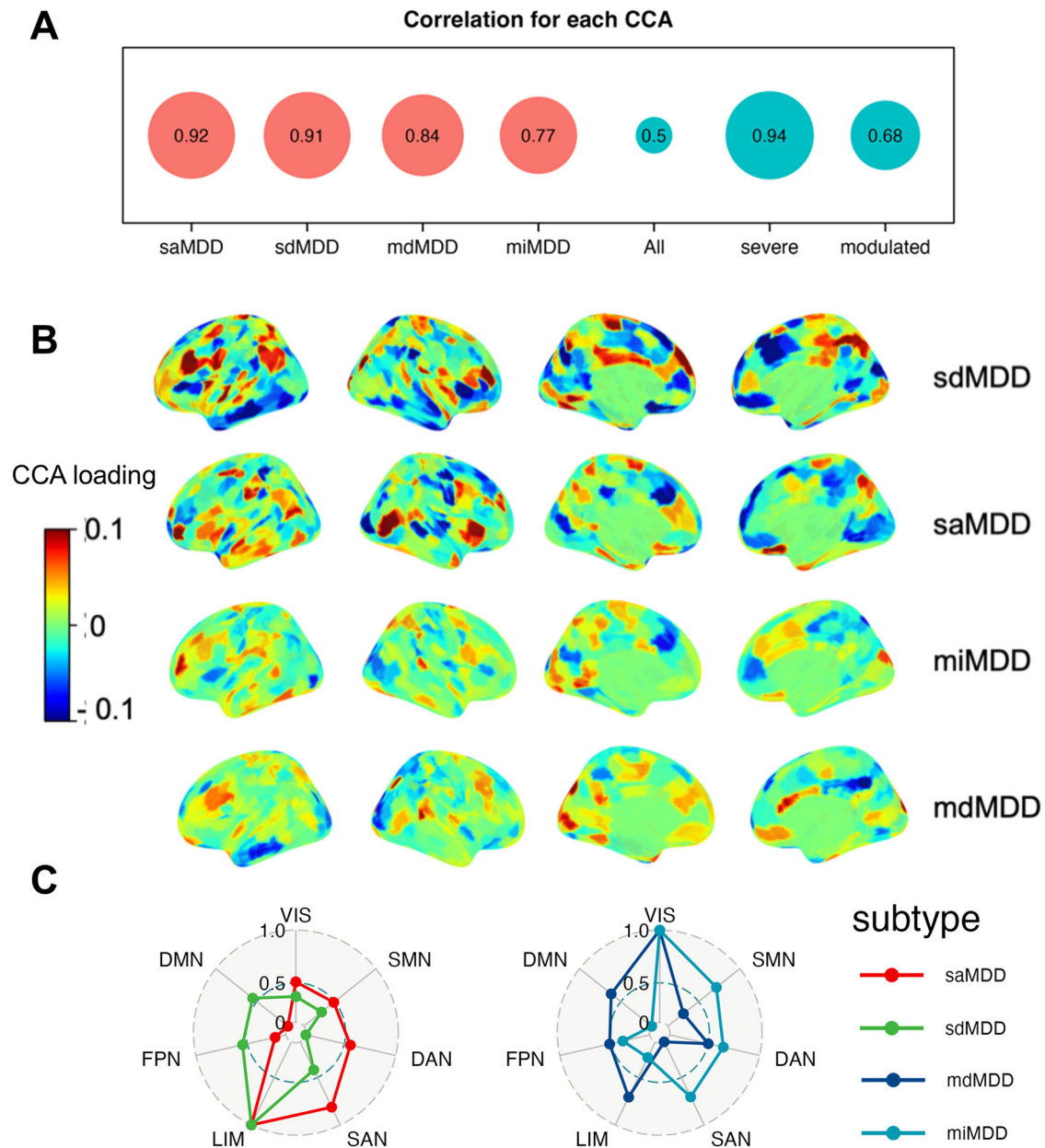


Fig. 3 Brain-symptom correlation map. A rCCA coefficient evaluated by different groups. The size of the circle represents the magnitude of the rCCA coefficient. **B** Illustration of the loading of each brain area to the first component of rCCA. **C** Network-level rCCA loading for different subtypes. The loading weights were normalized by the Max-min method across the networks for each subtype. rCCA regularized canonical correlation analysis.

9.56% in mdMDD, and 8.81% in miMDD subtypes. Coherence analysis between the PLS scores and HC-contrast TD maps revealed significant correlations across all subtypes. As illustrated in Fig. 4, the correlations were 0.23 for the saMDD, 0.24 for sdMDD, 0.3 for mdMDD, and 0.29 for miMDD subtypes (all $P < 0.001$)

As shown in Table 2 and Figs. S3 and S4, gene ontology enrichment analysis indicated that the genes highly ranked in the sdMDD subtype were enriched in biological processes linked to the regulation of metal ion transport. In the saMDD subtype, the gene rank was associated with sensory perception and synaptic signaling. In the miMDD subtype, the focus was primarily on the immune response, whereas in the mdMDD subtype, the genes were related to drug transport across the blood-brain barrier. Validation of the biological processes within the gene profiles of the saMDD and sdMDD subtypes was successfully achieved using the mirror-left-on-right gene expression extraction method

(Fig. S5). However, the replication attempts for mdMDD and miMDD did not yield successful results.

DISCUSSION

In the current study, we explored the distinct biological foundations of various depression subtypes. Using a latent profile clustering algorithm, we identified four depression subtypes based on their unique combinations of symptoms. First, by evaluating TD on fMRI data, we observed significant dynamic alterations in attention and limbic network for HC contrasts and among the subtypes. Additionally, by estimating the correlation between brain dynamics and symptoms by rCCA, we identified a distinct brain-symptom pattern across MDD subtypes. Specifically, we observed that the transmodal networks demonstrated a higher contribution to the rCCA component in the sdMDD and saMDD

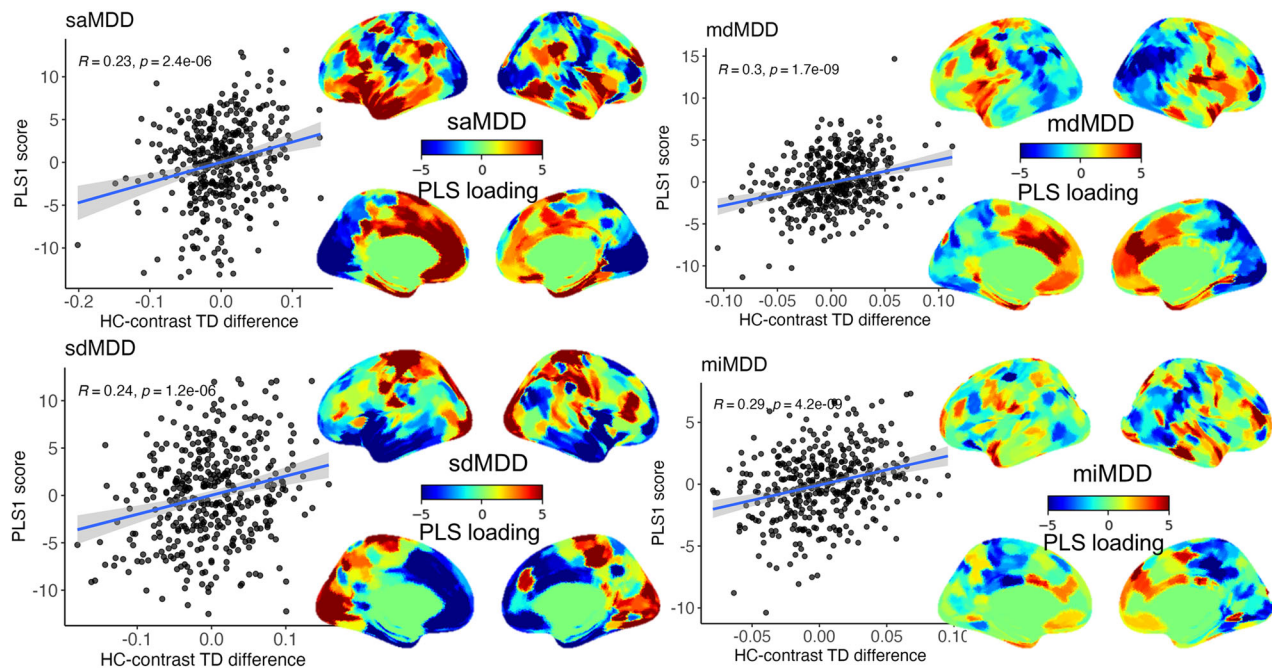


Fig. 4 Regional contribution to the PLS component and coherence to dynamic alteration. The dot plots show the Pearson's correlation between the PLS score of the brain regions (y-axis) and the time delay difference relative to HC (x-axis). The brain maps illustrate the spatial distribution of the PLS scores across the brain. Each of the four panels corresponds to a distinct subtype of MDD. PLS parietal least square, HC healthy control, MDD major depressive disorder.

Table 2. Biological process in GO analysis.

Subtype	Biological Process	P-value
sdMDD	Regulation of metal ion transport	8.2×10^{-6}
saMDD	System process	1.4×10^{-7}
	Cell-cell signaling	1.7×10^{-7}
	Anterograde trans-synaptic signaling	1.7×10^{-7}
	Chemical synaptic transmission	1.7×10^{-7}
	G protein-coupled receptor signaling pathway	2.8×10^{-7}
	Synaptic signaling	3.7×10^{-7}
	Sensory perception	5.0×10^{-7}
	CDP-diacylglycerol biosynthetic process	7.1×10^{-7}
	CDP-diacylglycerol metabolic process	7.1×10^{-7}
	Trans-synaptic signaling	8.5×10^{-7}
	Nervous system process	9.8×10^{-7}
miMDD	Inflammatory response	2.3×10^{-6}
	Immune response	2.5×10^{-10}
	Innate immune response	2.9×10^{-7}
	Detection of chemical stimulus	3.9×10^{-6}
	Detection of chemical stimulus involved in sensory perception of smell	4.2×10^{-7}
	Positive regulation of humoral immune response	6.7×10^{-6}
	Defense response	6.9×10^{-10}
mdMDD	Drug transport across blood-brain barrier	2.4×10^{-6}

CDP cytidine diphosphate.

subtypes, whereas the unimodal brain networks played a more significant role in the mdMDD and miMDD subtypes. Last, by performing gene enrichment analysis, we uncovered distinct gene ontologies in biological processes linked to brain-symptom correlation. Specifically, we associated gene rankings in the saMDD subtype with synaptic signaling transformation, those in the sdMDD subtype with mental ion transportation, those in the miMDD subtype with immune reactions, and those in the mdMDD subtype with substance exchange in the blood-brain barrier.

MDD subtypes and brain dynamic aberrance

Based on symptom combinations, we identified four subtypes, primarily focusing on depression, anxiety, and insomnia as their signature symptoms. The number of identified subtypes falls within the range reported in previous studies, which typically identified data-driven categories ranging from two to five [47]. Moreover, the manner of this symptom-based classification closely aligns with clinical practice [48]. For instance, anxious MDD, a subtype increasingly recognized in clinical practice [49, 50], is often observed in data-driven classification [51], and miMDD, a subtype characterized by significant insomnia, has also been identified in large-scale data research [48].

The brain dynamic patterns among the MDD subtypes exhibited distinct aberrations. Specifically, the miMDD subtype showed disrupted dynamics in the attention network, aligning with findings from a previous study [52]. The dorsal attention network, responsible for attentional states and reorienting responses [53], has been found to be compromised in sleep-related health issues [54]. Furthermore, we observed an early time delay in the dynamics of the limbic network in the sdMDD subtype compared with the two moderate subtypes. The limbic network, which is composed of the brain areas of the orbitofrontal cortex, subgenual anterior cingulate cortex, and temporal pole

gyrus, is a pivotal hub in neural circuits for emotion [55], as well as a treatment target for MDD [56, 57]. Moreover, a recent study revealed that dysfunction in the limbic network is correlated with the severity of anhedonia in patients with MDD rather than anxiety [58]. These studies suggest a link between the limbic network and the pathophysiology of MDD and an association between specific symptoms and this network.

Variance in brain dynamic and symptom correlations across subtypes

The correlation between brain dynamics and symptoms revealed distinct patterns across the four MDD subtypes. Notably, the CCA coefficient increased from 0.5 in all MDD patients to 0.7 to 0.9 when evaluated within relatively homogeneous subtypes. This finding underscores that MDD heterogeneity can disrupt the association between symptoms and brain dynamics, potentially undermining the statistical model's power [59].

Network-specific rCCA loadings highlighted the differential contributions of various networks to the brain-symptom correlation. Specifically, dynamics in the limbic network had the highest loading in two severe subtypes, which corresponded to our HC-contrast results, suggesting the association of dysfunction in the limbic network with depression severity. Nevertheless, dynamics in the visual network contributed the most to the correlation in the two moderated subtypes. Dysfunction of the visual network has been documented in patients with MDD [60], which may be related to the disruption of the GABA system within the visual cortex [61]. The shift in the highest loading network between severe and moderate MDD subtypes may reflect alterations in the primary impaired neural circuits as the disorder progresses.

We also observed that the transmodal network made the predominate contribution to the rCCA component in the sdMDD and mdMDD subtypes, whereas the unimodal network made the predominate contribution in the saMDD and miMDD subtypes. The transmodal network, including the default mode network and frontoparietal network, is involved in self-referential processing and cognitive controls [62, 63]. In the context of MDD, dysfunction of the transmodal network may relate to the symptoms of self-blame and self-criticism [63], as well as cognitive impairment [64]. The unimodal network, including the primary sensory networks, is involved in processing visceral sensory input and movement control [65]. Therefore, dysfunction of the unimodal network may relate to somatic symptomatology and psychomotricity [66].

Gene profiles indicate potential pathology of subtypes

Gene enrichment analysis has uncovered unique biological processes that underlie the gene profiles of different symptom combinations. For sdMDD, the biological process has been tied to mental ion transport, potentially connecting it to ferroptosis, a form of cell death triggered by an imbalance in reactive oxygen species, accompanied by significant iron accumulation [67]. Recent research has established a link between ferroptosis and depression [68, 69], primarily occurring as stress-induced depression [70]. This suggests that ferroptosis may affect resilience and feelings of hopelessness [71], which are the core symptoms of clinical depression.

The gene profile of saMDD, characterized by anxiety and psychomotor syndromes, has been associated with chemical synaptic transmission. Previous research has proposed that disruptions in the dopamine and 5-HT pathways might lead to psychomotor disturbances [72], aligning with our findings and possibly explaining why antidepressants are effective in treating anxious MDD [9].

The gene profile observed in miMDD has been found to be closely related to immune responses. The connection between neuroinflammation and depression is widely acknowledged [73, 74]. An excessive immune reaction in the brain, often triggered by a polarity shift in microglia, disrupts the production and transport of neurotransmitters, ultimately leading to depressive symptoms [74]. Additionally, disrupted sleep patterns can exacerbate the

accumulation of inflammatory cytokines [75]. Our findings indicate that patients with miMDD might derive significant benefit from a combined anti-inflammatory treatment approach.

The gene profile observed in mdMDD has been found to be associated with drug transport across the blood-brain barrier (BBB). Alterations in BBB permeability have been documented in psychotic disorders [76]. Postmortem studies indicate dysfunction in endothelial-associated tight junction components in the brains of patients with psychosis [77]. Malfunction of the BBB is linked to disturbances in the transport of various endogenous substances, including metabolites, hormones, lipids, and toxins, that can lead to imbalances in brain homeostasis and adverse health outcomes [78].

There are several limitations to our study that should be acknowledged. First, subtype identification relied on the HAMD score, which, despite its reliable psychometric properties, is subject to evaluator expertise. The "brain-first" approach, which categorizes subtypes based on neurobiological measure deviations [79], should be corroborated with findings from the "symptom-first" approach. Second, we derived the gene expression data from only six healthy donors with lateralization sampling bias, which potentially limits their representativeness to the psychosis population. Deriving gene expression data from the patients could enhance understanding of the transcriptome's role in brain dynamics and symptom correlation. Third, as this was a cross-sectional study lacking post-intervention data, we could not determine the appropriateness of specific therapies for different patient subgroups. Future research incorporating treatment responses for MDD subtypes would be beneficial for the development of individualized treatment plans tailored to the various subtypes of MDD.

DATA AVAILABILITY

The extracted fMRI signals, the time delay projection maps for each patient and the codes used in the present study are freely available on an online repository (https://github.com/QunjunLIANG/subtype_MDD). The raw images used in the present study are available from the corresponding author upon reasonable request.

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AUTHOR CONTRIBUTIONS

Referring to the Contributor Roles Taxonomy (CRediT) authorship contribution guidance, we made the following declaration: QL: Conceptualization, Writing – original draft, Writing – review & editing, Software, Formal analysis, Visualization. ZZ: Resources, Investigation, Writing – review & editing, Validation. SC: Data Curation, Methodology. SL: Writing – review & editing, Validation. XL: Writing – review & editing. YL: Investigation, Data Curation. YZ: Resources, Investigation. BP: Investigation, Writing – review & editing. GH: Funding acquisition, Project administration. YQ: Supervision, Funding acquisition, Writing – Review & Editing.

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ETHICS

The authors hereby confirm that all methodologies employed in this study adhere to the ethical guidelines set forth by the relevant national and institutional committees on human research, as well as align with the principles of the Helsinki Declaration. The present study was conducted under the approval of the Institutional Review Board of Shenzhen Kangning Hospital (NO.2020-K005–1). All involved individuals provided their written informed consent, ensuring their voluntary and informed participation in the study.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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